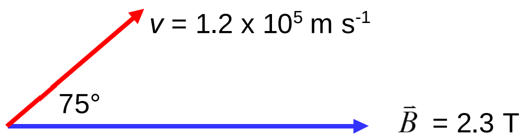
	$= 2 \times 10$ m
4(a)	From defining equation of Magnetic Flux Density, $F_B = BIL$ $B = \dfrac{F_B}{IL}$ $[B] = \dfrac{[F_B]}{[I][L]}$ $[B] = \dfrac{\text{kg m s}^{-2}}{\text{A m}}$ $[B] = \text{kg A}^{-1}\text{s}^{-2}$
4(b)(i)	Wire A experiences an electromagnetic force due to the interaction of the current flowing in the wire with the component of the external magnetic field that is perpendicular to the wire.
4(b)(ii)	It is not possible for wire B to experience an electromagnetic force as it is parallel to the external magnetic field, and there is no component of magentic field that is perpendicular to it for interaction to occur.
4(c)(i)	 $F_B = Bqv\sin\theta$ $F_B = (2.3)(1.6\times10^{-19})(1.2\times10^{5})\sin(75)$ $F_B = 4.265$ $F_B = 4.3$ N
4(c)(ii)	The electromagnetic force does no work on the electron. Work is defined as the product of force and the displacement in the direction of the force. As the force acts perpendicular to the motion (and hence displacement) of the electron, there is no work done since there is no component of the force that is parallel to the displacement.
5(a)	